COMPUTER SCIENCE - PYTHON (083)

CLASS XI - PROGRAMMING PROJECT / PRACTICAL ASSIGNMENT - SET D

Chapter	Dictionaries in Python	**Project Title**	Product Inventory Management
Level	CBSE Class XI	**Main Concepts**	Dictionary, loops, conditions and list

PROJECT QUESTION

Write a Python program to create and manage a Product Dictionary. Use **Product Code** as the unique key. For each product, store **Product Name, Category, Price and Quantity** as the corresponding record. The program must perform the following operations:

1. Create / Add Products: Accept product details and add records to the dictionary. Continue adding products until the user chooses to stop. Do not allow duplicate Product Codes.

2. Display Products: Display the Product Code and complete details of every product stored in the dictionary.

3. Search by Product Code: Input a Product Code and display the complete product record if found; otherwise display a suitable message.

4. Update Price: Input the Product Code of an existing product and allow the user to change only its Price.

5. Delete Product: Input a Product Code and remove the corresponding product record after checking that it exists.

6. Low Stock List: Prepare a separate list containing the records of all products whose Quantity is less than 10, and display the list.

7. Products in Price Range: Display the details of all products whose Price is between Rs. 500 and Rs. 1000.

PROGRAM REQUIREMENTS

• Use a dictionary as the main data structure and Product Code as a unique key.

• Use loops and conditional statements appropriately. Do not allow duplicate Product Codes.

• Check whether a Product Code exists before searching, updating or deleting a record.

• Use only Class XI-level Python concepts. Avoid advanced concepts such as file handling, functions, classes or database connectivity.

• Keep the program readable, properly indented and supported by meaningful output messages.

SUGGESTED DATA REPRESENTATION

The following structure is only for guidance. Students may use another simple Class XI-level representation that fulfils all requirements.

Product Code (Key)	Product Record (Value)
101	Name: Keyboard, Category: Computer Accessory, Price: 750, Quantity: 15
102	Name: Mouse, Category: Computer Accessory, Price: 550, Quantity: 8

EXPECTED LEARNING OUTCOMES

The student should be able to create and traverse dictionaries; search, update and delete records; apply conditions to dictionary data; identify records satisfying quantity or price conditions; and prepare a list from selected dictionary records.

Submission: Write the complete Python program and test it with sufficient product records so that every operation can be demonstrated.